Transverse divergences of the four–kernel row

A complete enumeration of the singular regions of $\int d^2z\, d^2y'$

addendum to *The four–kernel row: z and y' integrated*

0. The answer

Yes — exactly one, and it is infrared, not ultraviolet.

There is **no short–distance (UV) divergence**: every coincidence limit of the four kernels is integrable in two dimensions.

There **is** a **logarithmic divergence from the large–transverse–separation region**, $|y'-z| \to \infty$ (equivalently $|z|, |y'| \to \infty$). Its coefficient is exactly $-\pi\delta^{mj}$ per master, i.e. -2π in the trace, and it is **pure trace** — the traceless part of the integrand angular–averages to zero at large radius.

The oscillating phase regulates it, cutting the radial log at $r_{\max} = 2e^{-\gamma_E}/|Q|$. So it is a *genuine* divergence only where $|Q| \to 0$. That happens in **one place only**: the z–integration, whose conjugate momentum is $Q = -\xi k$, at the soft endpoint $\xi = p^+/k^+ \to 0$. The y'–integration has $Q = (1+\xi)k$, which never vanishes.

Physically therefore the transverse divergence of this row **is not a UV divergence to be renormalised**. It is the transverse companion of the rapidity divergence: it appears only at $\xi \to 0$, where it multiplies the $1/\xi$ pole of the $-\delta_{jm}\delta_{ik'}/p^+$ term and produces the double logarithm ℓ_k^2. It belongs to the JIMWLK/BK evolution kernel, not to a counterterm.

1. Setting up the enumeration

Both transverse integrations reduce to the same master (§3 of the main note),

$$B^{mj}(Q,c) = \int d^2r\; e^{-iQ\cdot r}\, \frac{r^m}{r^2}\, \frac{(c-r)^j}{(c-r)^2},$$

with the dictionary

integration	r	c	Q	**spectators**
$\int d^2y'$	$y'-z$	$x'-z$	$(1+\xi)k$	$x-z,\, y-w$
$\int d^2z$	$z-y'$	$x-y'$	$-\xi k$	$x'-y',\, y-w$

A two–dimensional integral $\int d^2r = \int \rho\, d\rho \oint d\theta$ can only diverge where the *angular average* of the integrand behaves like ρ^{-2} (a logarithm) or worse. So the complete list of candidate regions is: $r \to 0$, $r \to c$, $r \to \infty$, and the degenerate case $c \to 0$ in which the two singular points collide. Take them in turn.

2. Short distance: $r \to 0$ and $r \to c$ — both finite

Near $r = 0$ the second kernel is regular and the integrand is $\frac{r^m}{r^2}\frac{c^j}{c^2} \sim \rho^{-1}$; near $r = c$ the roles swap. In neither case is there a ρ^{-2}. Numerically, $\rho^2 \times \langle \text{integrand} \rangle_\theta$ must vanish, and it does — like ρ^2, i.e. the integrand really is $O(\rho^{-1})$ with zero angular average of the leading piece:

```
  r -> 0  (y'->z  /  z->y')  rho= 1e-01  rho^2*<integrand> max = 6.627e-03
                             rho= 1e-02                         6.627e-05
                             rho= 1e-03                         6.627e-07
  r -> c  (x'->y' /  x->z)   rho= 1e-01  rho^2*<integrand> max = 6.627e-03
                             rho= 1e-02                         6.627e-05
                             rho= 1e-03                         6.627e-07
```

Three decades of ρ, three decades of decrease each time: exactly ρ^2. $\int_0 \rho\, d\rho\, \rho^{-1} = \int_0 d\rho$ is finite. The same power counting covers the other two kernels ($y-w$ is a spectator, $x-z$ and $x'-y'$ are the $c-r$ factors of the two integrations), and it covers *simultaneous* coincidences too, because no two kernels become singular at the same point.

> This is worth stating sharply, because it is the whole structural difference from the previous row. There, the light–cone vertex supplied a second $\frac{1}{(x-z)^2}$ on top of the kernel $\frac{(z-x)^m}{(z-x)^2}$, so the integrand went as ρ^{-2} at $z \to x$ and $\int \rho\, d\rho\, \rho^{-2}$ was log divergent — the UV log with the P_{gg} coefficient. Here the contracting bracket carries only δ's and *longitudinal* denominators $1/(p^+ + k^+)$, $1/p^+$, $1/k^+$; there is no transverse denominator to double anything up.

3. Large distance: the one divergence

At large $|r|$ the second kernel becomes $\frac{(c-r)^j}{(c-r)^2} \to -\frac{r^j}{r^2}$, so the integrand tends to $-\frac{r^m r^j}{r^4}$, whose angular average is $-\frac{\delta^{mj}}{2r^2}$ — nonzero. Numerically:

```
rho= 1e+01   rho^2*<integrand> = [[-0.500000 -0.000000] [-0.000000 -0.500000]]
rho= 1e+02                       [[-0.500000 -0.000000] [+0.000000 -0.500000]]
rho= 1e+03                       [[-0.500000 +0.000000] [-0.000000 -0.500000]]
rho= 1e+04                       [[-0.500000 +0.000000] [+0.000000 -0.500000]]
```

so that, with the phase switched off,

$$\int^R d^2 r\, \frac{r^m}{r^2} \frac{(c-r)^j}{(c-r)^2} \;\longrightarrow\; -\pi\delta^{mj} \int^R \frac{d\rho}{\rho} \;=\; -\pi\delta^{mj} \log R + \dots$$

The divergence is purely the trace part. The off–diagonal and traceless–diagonal entries above are zero to six decimals, so no traceless structure is log divergent.

The same statement algebraically

Since $2\, r\cdot(c-r) = -\big[r^2 + (c-r)^2 - c^2\big]$, the trace of the integrand is exactly

$$\frac{r\cdot(c-r)}{r^2(c-r)^2} = -\frac{1}{2}\left[\frac{1}{r^2} + \frac{1}{(c-r)^2} - \frac{c^2}{r^2(c-r)^2}\right],$$

(verified to 2×10^{-16}). The last term is the finite dipole–like integral; the first two each give $-\frac{1}{2}\int d^2r/r^2$ at large r, adding to the announced $-1/r^2$. Note that the first two terms are separately singular at short distance as well and the third cancels those — consistent with §2.

The coefficient, measured

Setting $Q=0$ and cutting the radius at R, the truncated trace should be $-2\pi \log R + \text{const}$:

R	$\delta_{mj}\int_{\lvert r\rvert<R}$	$d/d\log R$
10^2	-30.288482	—
10^3	-44.756051	-6.283185
10^4	-59.223620	-6.283185
10^5	-73.691189	-6.283185

and $-2\pi = -6.283185$ to every digit shown.

4. What regulates it, and when the regulator fails

Restoring the phase, the exact master gives (main note, §5)

$$\delta_{mj} B^{mj}(Q,c) \;=\; -2\pi \log \frac{2e^{-\gamma_E}}{|Q|\,|c|},$$

which is precisely the cut–off form of §3 with

$$\rho_{\min} = |c|, \qquad \rho_{\max} = \frac{2e^{-\gamma_E}}{|Q|}.$$

The oscillation of $e^{-iQ\cdot r}$ destroys the integrand beyond one wavelength; that is the whole content of the logarithm. Checked against the exact master:

$\lvert Q\rvert\lvert c\rvert$	$\delta_{mj}B^{mj}$	$-2\pi\log\frac{2e^{-\gamma}}{\lvert Q\rvert\lvert c\rvert}$
8.72×10^{-2}	-16.073013	-16.056091
8.72×10^{-3}	-30.523911	-30.523659
8.72×10^{-4}	-44.991232	-44.991228
8.72×10^{-5}	-59.458797	-59.458797

The agreement improves as $|Q||c|\to 0$ because the $O(|Q||c|)$ remainder $\mathcal{B}^{mj}$ dies. **The transverse integral diverges logarithmically as $|Q|\to 0$, and only then.**

Where $|Q|\to 0$ actually happens

- $\int d^2y'$**: safe.** $Q=(1+\xi)k$ and $\xi\geq 0$, so $|Q|\geq|k|$. The gluon transverse momentum is a fixed external scale; the log is $\log\frac{2e^{-\gamma}}{(1+\xi)|k||x'-z|}$, finite for every ξ.
- $\int d^2z$**: divergent at the soft end.** $Q=-\xi k$, so $|Q|=\xi|k|\to 0$ as $\xi\to 0$, and the log becomes $\log\frac{1}{\xi}$ plus a constant. This is the only genuine transverse divergence of the row.

5. The degenerate region $c\to 0$, and why it is harmless

The same formula shows a logarithm when $|c|\to 0$: the two singular points of one integration collide. This is *not* an independent divergence of the double integral, because c is built from the other integration variable:

$$\int d^2y' \text{ has } c=x'-z \quad\Rightarrow\quad c\to 0 \text{ at } z\to x',$$
$$\int d^2z \;\text{ has } c=x-y' \quad\Rightarrow\quad c\to 0 \text{ at } y'\to x.$$

In the outer integral this appears as $\log|c|$ (or, for the nested T_1 structure, $\log^2|c|$) against the measure $d^2c=\rho\,d\rho\,d\theta$, and $\int_0\rho\,d\rho\,\log^n\rho$ converges for every n. So the collinear collision of the two Wilson–line arguments is integrable.

The same applies to the nested large–distance region of T_1: there the outer z integrand is a kernel $\sim|z|^{-1}$ times the inner $\log|x'-z|$. The kernel's leading angular average vanishes, leaving $\int d\rho\,\log\rho/\rho\sim\frac{1}{2}\log^2$ — cut at $\rho\sim 1/(\xi|k|)$ like everything else, and multiplied by $1/(1+\xi)$, which has no pole.

6. Consequence: the divergence is a rapidity divergence

Put §4 together with the longitudinal structure. The three terms of the collapsed bracket carry $\frac{1}{1+\xi}$, $-\frac{1}{\xi}$ and -1. Only the middle one has a pole at $\xi\to 0$, and it is exactly the term whose z–integration is the pure trace $\delta_{mj}B^{mj}$ — the maximally divergent contraction. So the transverse log and the longitudinal pole multiply:

$$-\frac{1}{\xi}\Big[-2\pi\log\frac{2e^{-\gamma_E}}{\xi|k||c|}\Big]\quad\Longrightarrow\quad 2\pi\int_\lambda^\Xi\frac{d\xi}{\xi}\,\log\frac{1}{\xi}\;+\;\ldots$$

which is the double logarithm. With $\ell_k=\log(k^+/\Lambda)$, $\int_\lambda^1\frac{d\xi}{\xi}\,2\log\frac{1}{\xi}=\log^2\frac{1}{\lambda}$ exactly:

```
lambda      int_lambda^1 dxi/xi 2log(1/xi)      log^2(1/lambda)
1e-02               21.2076                         21.2076
1e-03               47.7171                         47.7171
1e-04               84.8304                         84.8304
```

Summary. The four–kernel row has no transverse ultraviolet divergence. Its single transverse divergence is infrared — the large–separation region, coefficient $-\pi\delta^{mj}$, pure trace — and it is regulated by the phase at $r\sim 1/|Q|$. It becomes a true divergence only in the z–integration at $\xi\to 0$, where it is inseparable from the $1/\xi$ rapidity pole. Its correct disposition is therefore absorption into the rapidity evolution of the Wilson–line correlators, not a transverse counterterm.

`zint/src/ir_check.py` — every number above, written to `zint/RESULTS_nlo2_ir.txt`. It uses the master `zint/src/nlo2_master.py` from the main note.